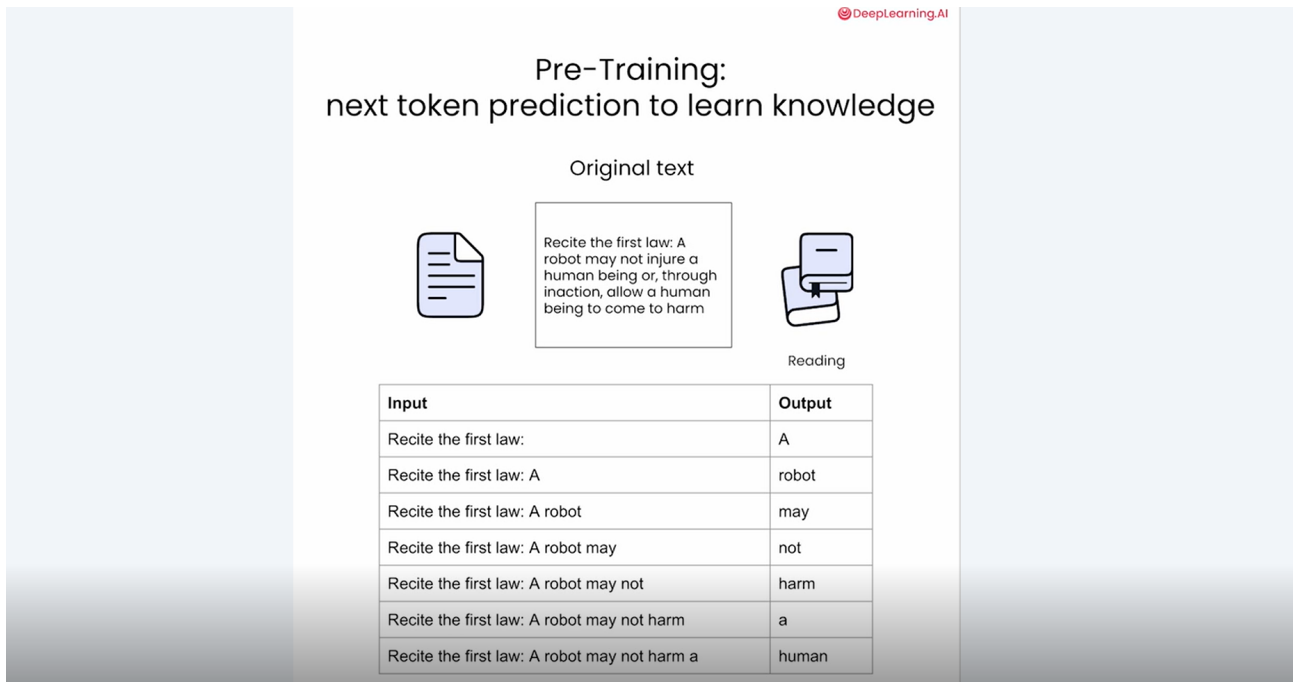


Pre-Training

Next-token prediction to learn knowledge



1. What is pre-training?

Pre-training is the first major training stage of a large language model (LLM). The model is exposed to a very large amount of text and learns general language patterns, facts, relationships, syntax, and context. The example in the image shows the common objective used for decoder-style language models: **next-token prediction**.

2. What does “next-token prediction” mean?

The model reads a sequence of tokens and tries to predict the token that comes next. For example, if the input is:

Recite the first law:

the correct next token is **A**. If the input becomes “Recite the first law: A”, the next token is **robot**. The process continues one token at a time.

3. How the table in the image works

Input (context)	Target / output
Recite the first law:	A
Recite the first law: A	robot
Recite the first law: A robot	may
Recite the first law: A robot may	not